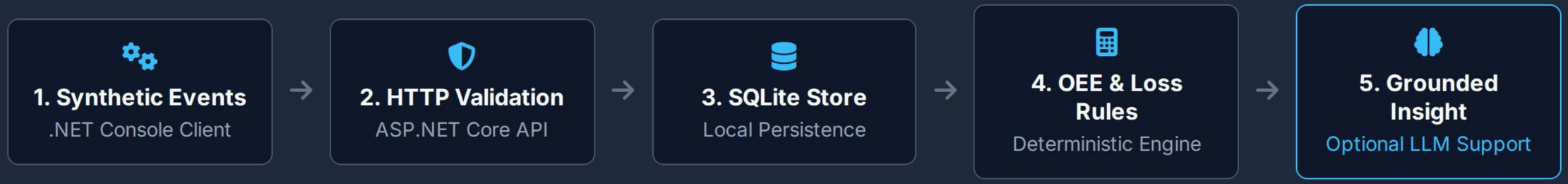
ARCHITECTURE SPECIFICATION Laptop-Scale Prototype

OEE & Loss Analysis System Architecture

Providing shift leads with real-time operational effectiveness visibility and deterministic loss breakdowns.

PROTOTYPE PIPELINE FLOW



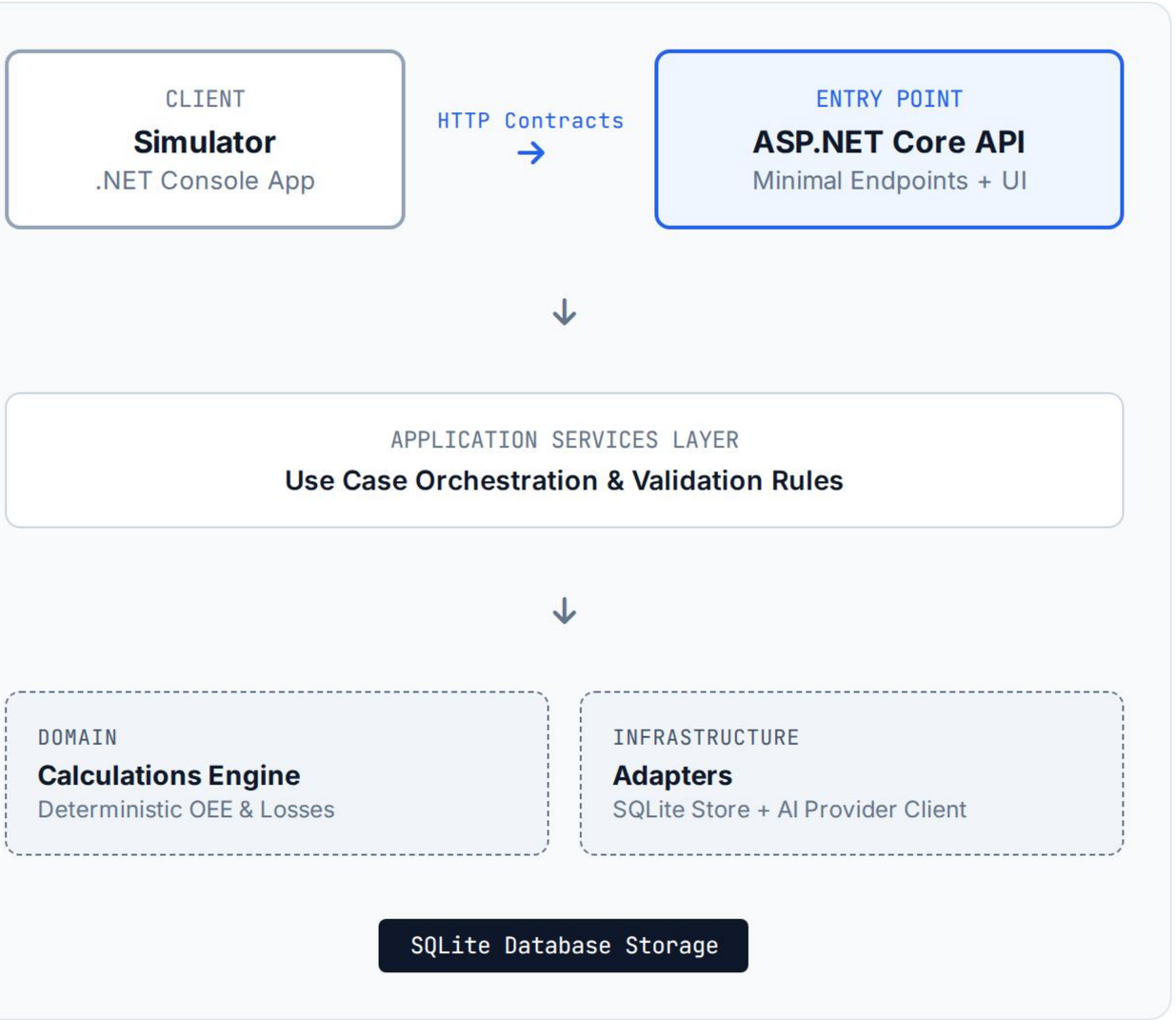
Single-Process Target: One API process, single SQLite DB file, one simulator client. No external brokers or clusters.

Developer Experience: Embedded browser frontend with integrated Swagger/OpenAPI interactive endpoints.

LAYERED DESIGN

Component Architecture & Boundaries

Explicit separation of concerns



API Host Layer

Composes services, exposes HTTP routes, serves static browser UI, and handles exceptions globally.

Application Layer

Orchestrates operations and declares core abstractions: `IProductionStore` and `IInsightGenerator`.

Domain Layer

Pure production entities and deterministic formulas. Completely free of external dependencies.

Infrastructure Layer

Implements persistence via SQLite and AI integrations (Ollama / OpenAI-compatible services).

Shared Contracts: The simulator references isolated DTO contract records directly rather than API internals.

SCHEMA & INGESTION

Data Model & Ingestion Pipeline

ProductionLine

Id: Stable String Identifier

DisplayName: Human Readable Title

IdealCycleTimeSeconds: Baseline Speed

ProductionShift

Id & LineId: Relational Mapping

StartUtc / EndUtc: Fixed Windows

PlannedProductionSeconds: Net Budget

ProductionEvent

Id: Client-generated GUID

Counts: Total / Good Production Units

Downtime: Start/End Interval & Reason

Simulator Workload

Targets first 2 lines by default. Provisions three 8-hour shifts per line. Submits 10 production + 5 downtime events per shift.

Idempotency & Retry Guarantee

Identical event GUID retries return existing record safely. Conflicting payload data with identical ID is rejected.

API Validation Gate

Enforces UTC boundary checks, non-negative unit counts, mandatory downtime reasons, and non-overlapping intervals.

DETERMINISTIC FORMULAS

Metrics & Loss Analysis Engine

API Calculated Post-Persistence

AVAILABILITY

Runtime / Planned Time

Downtime & changeovers reduce factor

PERFORMANCE

Ideal Time × Units / Runtime

Unclamped signal (>100% flags config error)

QUALITY

Good Units / Total Units

Evaluates yield purity

OVERALL OEE

A × P × Q

Final composite score

Planned Time Boundaries

Scheduled breaks are excluded up front. Downtime inside planned hours directly penalizes Availability.

Loss Aggregation Rules

Downtime events are grouped by reason, cumulative duration, and occurrence count for Pareto analysis.

Zero-Division Safeguards

Zero runtime or zero total unit output safely produces zero for applicable ratios without throwing runtime exceptions.

INTELLIGENCE & GOVERNANCE

AI Integration & Grounding Safeguards

Supported Providers

Mock Provider
Deterministic, offline template output for testing.

Ollama (Local Default)
Zero-latency local LLM evaluation on host machine.

OpenAI-Compatible
Standardized interface for approved remote endpoints.

Rule: API computes metrics first. Pre-calculated figures + evidence are passed to AI.

Validation & Boundary Rules

Strict Evidence Rejection
Application strictly rejects responses referencing missing or non-existent event IDs.

Prohibited Generative Claims
System prompts forbid inventing events, fabricating metrics, or confirming root causes.

Structured JSON Parsing
Normalizes summaries and recommendations, requiring verified event-ID evidence tags.

Human-in-the-Loop Position: AI output serves purely as reviewable decision support — never automated physical line control or authoritative math.

OPERATIONS & ROADMAP

Operations Baseline vs. Production Evolution

Calculation Tests Verified

Current Prototype Baseline

- **Tech Stack:** .NET 10, SQLite, ASP.NET Core API, embedded browser UI, and simulator.
- **Logging & Tracing:** Daily rolling file logs with trace IDs, HTTP status, execution timing, and exception stacks.
- **Automated Test Suite:** Deterministic tests covering calculations, loss handling, AI responses, and evidence validation.

Prototype Scope: No authentication, RBAC, DB migrations, high-availability, or regulated AI governance.